

A Packing of 9 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 9 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.499761. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.468863, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 9 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.499761 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 9$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.468863, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 9/8 = 2.500000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N9.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
249976157933993835506328682637003143723080611774086693149307140161803923
331883049470516518206136512257209212713330936139333099853181475163467921
1915194653475628458919216205754206928948964080419
```

denominator:

```
100000000000251933741189452491349537829289082145450421399593726927380075
893682000847036482176370647095901571245176120825874998954958977194534983
3446981075874972816331965669171198877316976035708
```

B Coordinates

9 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N9.txt`.

```
714645901501 407456795839 -93141873653 -348644333659 38258876080 -78217747571 18108671712 152602705432 -308833035657 -92770033470 389532404047 -508669776910
-76078149028 74266761121 -296039181941 214693585849 -329249748755 152866892254 120159427487 220582814237 07025463293 472370814251 -599647056895 -77850116592
16393870083 -179995219766 -11324930150 -115023772228 3052182793 493097505270 69939029858 -78345265075 201186428029 224472009686 -284101249510 -737974145423
283128219654 3330130894 -315006613080 287737959551 357185890708 328161972461 -493054341524 -263536579346 5781347000 -150753782182 -391506772742 164322047878
204094920030 -178486720475 475522004595 -186143724536 7030740134 -291812834131 -222846897036 -638440797399 0870707 -203052009569 -92447890382 -208325755203
-114535496393 105298357441 -646069143059 -494724883762 28557211 -246819471096 -134887883811 -361444442693 68622859 360430837020 -63375524691 319195438985
20969265683 612565043056 12042832204 568242364462 31273208354 -597287615641 181955150744 -4793667629 -269670116440 283704232262 284594948172 -100725596198
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249743682536 228405311594 331215157808 -118871701942 283186182692 -117775725152 -793116643619 390429098368 075524608 -581207043453 -398430619129 -90233601665
-386108205727 -1960132090 306950503980 -343277293892 78799868298 82945139677 -355779282387 380419275785 -539869621325 -352772822578 207730990142 -136623573748
-27276268440 -117514123098 -389441906907 -501609064636 6471010112 -562042850886 474104870648 286247274236 5972883146 -609767118297 233560959166 -369209339877
-666818225600 219341631253 -107388585529 522518976109 2554131155 -160682508629 -598093665303 -441295429303 643895759 42003721664 -352976822497 -562558896471
-124845893218 -876082707240 188835519533 296841694352 2627163906 -100371806515 -209549093354 364495969297 096077290 -298493607599 436805631355 63131402680
223566108171 61962400445 576704629719 -132650441352 28849815251 -597605316583 35455499884 -401589928756 62218636552 -425832430338 -204584854356 302984667208
159896778469 -97442664111 -594291800939 1002001743 339885675294 -386441387967 -384747500222 499840514521 972444901 128941553349 -144998264607 82555362821
375424947931 -4282548232 65636074901 299532180889 148858637773 -114053861793 -55160875263 8455994325 358519448726 330927284002 359078853735 -589783359361
202094569915 -295969999389 -111924718634 -75618963107 948475164 182203546353 305636078237 92298900129 -146060410006 -324407657346 -547366711537 -259010594929
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107881356521 -98744497185 -535203506078 -516536033303 823706631 123797177338 42116279456 78710906054 381727282267 48076594394 -731480260728 446204292498
-95990308618 923172698922 82014040764 -47840485551 -41176790958 -362990919703 331984237477 123070805482 2798364824 210165960440 -222356864630 16519647540
-169771155842 7100731294 420080024753 50327472618 270193251336 -234738022894 416509683594 -200147160945 9619486995 -373591569367 -189274369167 410080026133
-767938446403 -123139701012 -353010907901 679718207228 195822904 89226064289 -622859411558 -440784287814 1539240951 -250612694437 108148710194 122938178752
23826103166 239895667215 -355249087566 -85512727892 90957064646 -113800031661 259310621191 -481884759793 0407939336 460450842967 315587095946 496534960524
441930136456 -67494800502 133412938299 -382156912590 497608284 372610694469 483492550687 -231751809118 47587053494 -398982416423 469522050193 417989819078
270945735835 158633578251 -326945342820 45371741323 36832555193 92180598494 -40603049378 358340281319 -459478294118 -182261403714 -83938167395 287302331112
1601115151 -183742390024 230203742407 293748834442 -27962881764 772382783894 144875714174 -99677545083 -340966028926 -61312649114 -179961404637 -246614123163
-302178236659 41910960457 319153614071 -527488947792 6017952070 181623282434 54165459191 565723698234 211312791135 -585628823744 -92169565080 -229620983128
```